

Adroit Technologies

Protocol Driver

Name	Modbus Plus MBX API
DLL	MODBUSP



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1. Overview

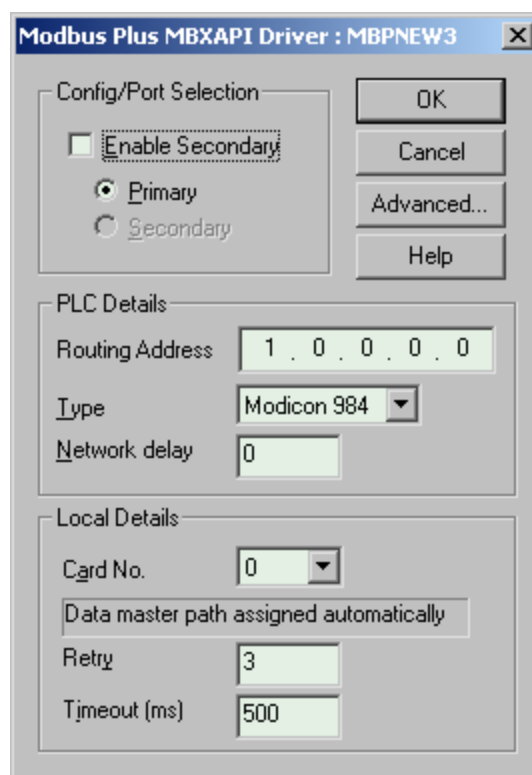
Adroit supports communication with the following PLCs: Modicon 84, 884, 184, 384, 484, 584 , 984 and Moscad via the SA85 adapter and accompanying Cyberlogic device drivers. This driver does not support peer cop.

2. Wiring

Network dependent.

3. Device Configuration

3.1. Device Configuration



3.1.1. Port Selection

Enable Secondary – Enables secondary configuration for dual redundant Controllers. This setting is disabled for this driver.

Primary / Secondary – Select the primary or secondary button depending on which connection you wish to configure.

3.1.2. PLC Details

Routing Address - Routing path of up to 5 levels, e.g. 25.2.200.0.0 will address multiplexer node 25, port 2 slave device 200.

PLC Type - Applicable Modicon PLC types (as detailed above)

Network delay – Network inter-PLC quiet time (see below).

3.1.3. Local Details

Card No. - The SA85 adapter number on your machine. (0 corresponds to the first card)

Retry - The number of times that Adroit will try to gain communication with a PLC before failing a transaction.

Timeout - The time that Adroit will wait for a valid response from a PLC before failing a transaction (Read or Write).

3.2. Data Master Path Note:

The Data master path is a logical path or connection resource, paths are assigned automatically. Any configured devices on the same route now automatically use a new path as well. This means that 2 devices on separate "paths" may communicate simultaneously for better speed.

3.3. Network Delay Note

Applies to RS-232/422 gateway use only.

The network delay is an optional instituted delay, inserted before a different station is polled on the same Data master-path. Using a value of zero will disable this feature. The network delay or quiet time is needed to allow a slave to detect the end of the communication cycle.

A Modbus slave interprets a 2.5 character time delay at the end of a message as a signal that communication has been broken off, and thus allows communication to be initiated by the master to a new slave. By default, the Adroit driver does not implement this delay or network quiet time.

To calculate the necessary time use the following formula, we have used 5 characters instead of 2.5 as this is always good engineering practice. 11 bits per byte have been used.

$1/\text{Baud} \times 11 \times 5$.

e.g. For a baud-rate of 2400, $1/2400 \times 11 \times 5 = 0.022916$. or 23 ms.

For a baud-rate of 19200, $1/19200 \times 11 \times 5 = 0.002863$. or 3 ms.

Devices using separate Data master paths will not wait using this network quiet time, so that all 422 communications must take place on the same data master path.

4. Supported Addresses

Modicon PLC data address types supported by the MODBUS Adroit Protocol driver

Device		Min Range	Max Range	Plc Data Type	Boolean	Integer	Real	String
Outputs	1	00001	09999	Digital	00001			
	1			Decimal				
	2			Signed Integer				
	3			BCD16				
Inputs	4	10001	19999	Digital	10001			
	4			Decimal				
	5			Signed Integer				
	6			BCD16				
I/O Registers	7	30001	39999	Digital	30001.1			
	7			Decimal		30001	30001	
	8			Signed Integer		30001 I	30001 I	
	9			BCD16		30001 B	30001 B	
Holding Registers	10	40001	49999	Digital	40001.1			
	10			Decimal		40001	40001	
	11			Signed Integer		40001 I	40001 I	
	12			BCD16		40001 B	40001 B	
	13			Long Decimal		40001 L	40001 L	
	14			Float			40001 F	
	15			String				40001 520



PLEASE NOTE

Integer and Real writes are not supported to coils 00001 through to 19999 . These are read only mappings. The same is true of Boolean mappings to coils 10001 through to 19999.

The Inputs Digital BOOLEAN is Read only.

The maximum number of items (Digital Bits or 16 Bit Registers) that can be communicated for the different Modicon PLC's is shown below:

	84	884	184	384	484	584	984	Moscad
00001	64	2000	800	800	512	2000	2000	256
10001	64	2000	800	800	512	2000	2000	256
30001	4	125	100	100	32	125	125	16
40001	32	125	100	100	254	125	125	16

5. Protocol Definition and Considerations

Datascope text now appears on "ASCII" mode, each transaction is preceded by cursory explanatory text in WHITE. Toggle HEX/ASCII modes using the "a" key on the keyboard or by selecting ascii mode in the driver monitor launcher.

This display is only visible for Adroit 5.0 users onwards. Adroit 4.2 systems require patches:
 drvmon.exe
 pdrvmon.exe

5.1. Coils 00001- 09999

Read request:

Slave Address	Function	Start Addr Hi	Start Addr Lo	No Points Hi	No Points Lo	CRC
2	01	00	13	00	10	xx

Read response:

Slave Address	Function	Byte Count	Data Hi	Data Lo	CRC
2	01	02	02	0F	xx

Write request:

Slave Address	Function	Coil Addr Hi	Coil Addr Lo	Force Data Hi	Force Data Lo	CRC
2	05	00	13	FF	00	xx

Write response:

Slave Address	Function	Coil Addr Hi	Coil Addr Lo	Force Data Hi	Force Data Lo	CRC
2	05	00	13	FF	00	xx

5.2. Inputs 10001- 19999

Read request:

Slave Address	Function	Start Addr Hi	Start Addr Lo	No Points Hi	No Points Lo	CRC
2	02	00	13	00	10	xx

Read response:

Slave Address	Function	Byte Count	Data Hi	Data Lo	CRC
2	02	02	02	0F	xx

Write request:

Not supported

5.3. Input Registers 30001- 39999

Read request:

Slave Address	Function	Start Addr Hi	Start Addr Lo	No Points Hi	No Points Lo	CRC
2	04	00	13	00	1	xx

Read response:

Slave Address	Function	Byte Count	Data Hi	Data Lo	CRC
2	04	02	02	0F	xx

Write request:

Not supported

5.4. Holding Registers 40001- 49999

Read request:

Slave Address	Function	Start Addr Hi	Start Addr Lo	No Points Hi	No Points Lo	CRC
2	03	00	13	00	1	xx

Read response:

Slave Address	Function	Byte Count	Data Hi	Data Lo	CRC
2	03	02	02	0F	xx

Write request:

Slave Address	Function	Start Addr Hi	Start Addr Lo	No Registers Hi	No Registers Lo	Byte Count
2	10	00	13	FF	02	xx

Data Hi	Data Lo	Data Hi	Data Lo	CRC
2	10	00	13	xx

Write response:

Slave Address	Function	Start Addr Hi	Start Addr Lo	No Registers Hi	No Registers Lo	CRC
2	10	00	13	00	02	xx

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

6.3. Troubleshooting

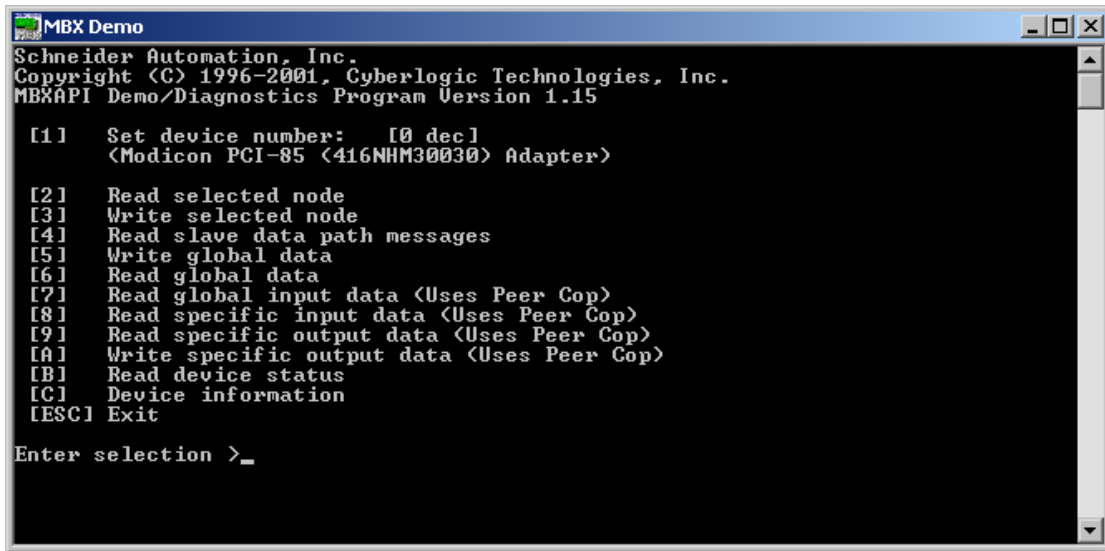
6.3.1. After installing the Adroit driver, I do not see it in setup!

If you have not installed Cyberlogic drivers for the adapter, the Adroit driver will not be shown in setup. Adroit setup version 5.0.4.0 onwards will also alert you that the driver modbus.dll failed to initialize correctly. Driver Version 18 onwards utilizes the Cyberlogic MBXAPI, when installed mbxapi.dll will be placed in the %WINDOWS%\SYSTEM32 folder, this driver supersedes the netlib.dll interface.

6.3.2. To verify card operation and network paths:

Run the mbpdemo program either from the diagnostics tab in the driver setup program as shown in the Device Driver Installation Notes section.

Or Click "Start/Programs/WinConX/MBX Driver/MBX Demo"



```

MBX Demo
Schneider Automation, Inc.
Copyright (C) 1996-2001, Cyberlogic Technologies, Inc.
MBXAPI Demo/Diagnostics Program Version 1.15

[1] Set device number: [0 dec]
    <Modicon PCI-85 <416NHM30030> Adapter>

[2] Read selected node
[3] Write selected node
[4] Read slave data path messages
[5] Write global data
[6] Read global data
[7] Read global input data <Uses Peer Cop>
[8] Read specific input data <Uses Peer Cop>
[9] Read specific output data <Uses Peer Cop>
[A] Write specific output data <Uses Peer Cop>
[B] Read device status
[C] Device information
[ESC] Exit

Enter selection >_

```

Typical steps to follow:

- Select option [2]
- Type in your route, example 1.0.0.0.0 followed by <ENTER>
- Type in a register number example 40001 followed by <ENTER>
- Type in the number of consecutive registers example 10 followed by <ENTER>

The display will continuously update showing the desired register values in HEX and decimal formats. To stop and return to the main menu, press the <ESC> key.

6.3.3. The driver gives an "Access denied:" error message after 2 hours of running, and never recovers.

You need to activate your license, see the packaging for details.

7. Revisions

7.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1		K. Robinson	Changes to format, updated image resources.	D. Jordaan
2.0				

7.2. DLL Revision

Rev.	Date	Changes
	13-Jul-1998	Added configuration option for RS-422 network connection for network delays or latency. Also added device driver configuration screen-shots.
	5-Apr-2005	Driver was re-developed using MBXAPI drivers: Netlib, the netbios interface is no longer used, also the device-driver screen shots are now replaced by new ones. Added web links section.
31	28-Sep-2018	Perfmon counters were not visible when running the Agent Server as a service.

